O-F-501b
MARCH 25, 1958

SUPERSEDING
Int. Fed. Spec. O–F–00501a (GSA–FSS)
June 12, 1957, and
Fed. Spec. O–F–501
February 20, 1947

FEDERAL SPECIFICATION

FLUX, CASTING, DENTAL, (REDUCING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 *This specification covers flux used in* dental prosthetic procedures for reducing metal alloys.

1.2 Classification.

1.2.1 *Type.*—This specification covers one type of dental casting flux.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

PPP–B–636—Boxes, Fiber.

Interim Federal Specification:

PPP–M–00186 (ARMY-MED) — Medicinal Products and Clinical Laboratory Reagents; General Specification for Containers, Packaging and Packing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—The reducing flux shall consist essentially of sodium pyroborate, an ac-

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tive reducing agent, and other ingredients so selected as to produce flux conforming to the requirements of this specification. The flux shall not contain hydrated borax.

3.2 Characteristics. — The reducing flux shall be such that when used in melting gold alloys for casting it will melt at a lower temperature than the metal. It shall not volatilize excessively at a temperature of 2,000 degrees F., and shall be fluid enough to cover the metal, yet be viscous enough that it will not flow off the metal. The molten flux shall be strongly reducing for oxides of the base metals present in casting gold alloys.

3.3 Form.—The reducing flux shall be supplied in the form of fine powder, free from lumps and ready for use.

3.4 Fineness.—Ninety percent of the flux powder shall pass through a No. 80 standard mesh sieve.

3.5 Workmanship.—Workmanship shall be first class throughout. The powder shall be free from any impurities that would interfere with the intended purpose.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling. — Sampling shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105. In addition, samples may be requested by the procuring activity for testing at Government laboratory.

4.2 Inspection.—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at a point of production or at point of delivery, as specified (see 6.1). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

4.3 Tests.—Two pennyweights of casting gold alloy shall be heated to a casting temperature and a small amount of flux applied. It shall reduce the oxide formed leaving the nugget bright and shiny.

5. PREPARATION FOR DELIVERY

(The levels of preservation, packaging and packing used are defined in Fed. Std. No. 102.)

5.1 Packaging.

5.1.1 *Unit of issue.*—One (1) ounce of flux, as specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE I.—*Packaging quantities*

Federal Stock Number	Unit package	Intermediate package	Shipping container
6520–531–1350	1 unit	12 units	288 units

5.1.2.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be packaged in a jar conforming to Inter
eral Specification PPP–M–00186 (ARMY-MED), group A, class 1, type e, style 4, grade 1 or 2, closure B, with inner or outer seal.

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5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636, type I, class 1. Carton design shall include full-height partitions fabricated of the same strength material as body of carton, so arranged that each jar is securely held in an individual cell. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 *Level C.*—Unless otherwise specified, commercial packages are acceptable.

5.2 **Packing.**

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table IV.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.*—Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table II.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Unit package.* — Each jar shall be marked with stock number, item identification, unit, quantity, and name of contractor.

5.3.2 *Intermediate package.*—Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.3 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Inspection required (see 4.2).
(*c*) Place of final inspection (see 4.2).
(*d*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Purchasers should exercise any desired options offered herein. (See 5.1, 5.2, and 5.3.)

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the title of the specification, or which are commercially available,

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but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.5 This specification covers the following item appearing in the Armer Services Medical Stock List:

6520–531–1350 — FLUX, CASTING, DENTAL, Reducing, 1 oz.

6.6 Transportation description.—Transportation description applicable to this item is:

Solvents.
Carload minimum weight 36,000 pounds.
Motor volume minimum weight 36,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

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Copies of this specification may be purchased for 5 cents each.

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